4+1 Architectural View

Mobile Repair Platform

Smart System for Connecting Mobile Repair Technicians to Customers

1. Logical View

The logical view captures the static structure of the Mobile Repair Platform regarding domain entities and their relationships. It supports the specification of services, behaviors, and responsibilities required for managing repair requests and technician interactions. UML Class Diagrams and ER Diagrams are commonly used to express this view.

- **Purpose:** To describe the functional structure of the mobile repair platform using object-oriented concepts.
- **Key Stakeholders:** Software designers, application developers.
- **Concerns:**
  - Customer, Technician, Admin, Repair Request, Mobile Device, Payment, Rating, Spare Parts.
  - Relationships among system entities.
  - Functional decomposition and object interactions.
- **What is Irrelevant:**
  - Hardware configuration.
  - Physical deployment details.

2. Process View

The process view describes the dynamic behavior of the system and how different components interact during runtime to process repair requests.

- **Purpose:** To model the execution flow of repair services and communications.
- **Key Stakeholders:** Software architects, system engineers.
- **Concerns:**
  - Customer submits a repair request.
  - Technician receives and accepts the request.
  - System updates repair status.
  - Online payment processing.
  - Notification service.
- **What is Irrelevant:**
  - Source code organization.
  - Database schema details.

3. Development View

The development view focuses on how the project is organized for implementation. It represents the structure of the source code in terms of packages or components and their dependencies.

- **Purpose:** To describe the software modules and package structure.
- **Key Stakeholders:** Developers, software architects.
- **Concerns:**
  - Authentication Module
  - Customer Module
  - Technician Module
  - Repair Request Module
  - Payment Module
  - Notification Module
  - Admin Module
- **What is Irrelevant:**
  - Runtime execution flow.
  - Hardware infrastructure.

4. Physical View

The physical view explains how the system is deployed on hardware and network resources.

- **Purpose:** To describe the deployment environment.
- **Key Stakeholders:** Network administrators, system administrators.
- **Concerns:**
  - Mobile Application
  - Web Application
  - Application Server
  - Database Server
  - Internet Connection
- **What is Irrelevant:**
  - Internal business logic.
  - Class relationships.

5. Scenarios (+1)

1. Customer → Login → Submit Repair Request → Technician Accepts → Repair Completed → Customer Pays → Customer Rates Technician
2. Technician → Login → View Requests → Accept Request → Update Repair Status → Finish Repair
3. Admin → Login → Manage Users → Manage Technicians → Monitor Repair Requests → Generate Reports

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